

|           |          |                                                                                                                                |
|-----------|----------|--------------------------------------------------------------------------------------------------------------------------------|
|           |          | $I_T \propto A_0^2 = \frac{1}{\sin^2 \theta}$                                                                                  |
| <b>18</b> | <b>B</b> | $\frac{\lambda}{2} = 0.49 - 0.15$ $\lambda = 0.68 \text{ m}$<br>$v = f \lambda$ $330 = f (0.68)$ $f = 490 \text{ Hz (2 s.f.)}$ |
| <b>19</b> | <b>D</b> | The glass block has higher refractive index and the speed of wave and hence the                                                |